



Topological Sort, Homework 10

1 Directed Graphs

1. Let G be a DAG, prove that G^T is also a DAG.
2. Proof the following claim¹

Claim 1. Let G be a directed graph. Define the relation $v \sim u$ to hold if u and v are on the same strong connected component. Then:

- (a) $\sim$ is an equivalence relation.
- (b) Define the graph $G' = (V', E')$ such that V' are the equivalence class respecting to $\sim$ and $\{u, v\} \in E'$ if there is a directed path in G from a vertex in u into a vertex in v . Then G' is a DAG.

2 Topological Sort

1. Algorithm 1 presents a variation of topological sort where the final DFS iterates over the vertices in increasing post-order instead of decreasing as in the original version. Give an example of a **family** of directed graphs for which the algorithm fails to return the strongly connected components.

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1 pre, post  $\leftarrow$  DFS( $G^T$ )  
2  $V \leftarrow$  sorts  $V$  in order of post.  
3  $\pi \leftarrow$  DFS( $G$ )  
4 return  $\pi$ 
```

Algorithm 1: Wrong topological sort.

¹From the recitation.